$q_3 = q_2 a \rightarrow ①$

$q_2 = q_1 a + q_2 b + q_3 b \rightarrow (2)$

$q_1 = \epsilon + q_1 a + q_2 b \rightarrow ③$

$① \rightarrow q_3 = (q_1 a + q_2 b + q_3 b) a$

$= (q_1 aa + q_2 ba + q_3 ba) \rightarrow a$

$\cancel{= q_1 a + 2 q_2 ba}$

$(2)\ q_2 = q_1 a + q_2 b + q_3 b$

$= q_1 a + q_2 b + (q_2 a) b$

$= q_1 a + q_2 b + q_2 ab$

$q_2 = q_1 a + q_2 \underbrace{(b + ab)}$

R Q R P

$R = Q + RP$

$q_2 = (q_1 a)(b + ab)^* \ (5)$

3 $q_1 = \epsilon + q_1 a + q_2 b$

$= \cancel{\epsilon + q_1 a + (q_1 a + q_2 b +}$

$= \epsilon + q_1 a + ((q_1 a)(b + ab)^*) b$

$q_1 = \epsilon ((a + a(b + ab)^*) b)^*$

$q_1 = (a + a(b + ab)^* b)^* \rightarrow ⑥$

Final state = ◎ $q3 = q_2 a$

$= q_1 a(b + ab)^* a$

$= (a + a(b + ab)^* b)^*$

$a(b + ab)^* a$

$\overset{q3}{R} = Q + \overset{q3}{R} P$